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END SEMESTER EXAMINATION 2024

Name of the Course: B.Tech Petroleum Engineering

Name of the Course: Thermodynamics

Semester: III

Course Code: TPE-301

Time: Three Hours

MM: 100

Note:

- All questions are compulsory.
- Answer any two sub questions in each main question.
- Total marks for each main question is **twenty**
- Each Question carries **10 marks**

Q1		COs
a.	Discuss Kelvin Planck's statement of 2 nd law of thermodynamics? a) 1 st Maxwell's relations:----- b) Fugacity coefficient is the ratio of fugacity to pressure (True/False) c) The heat capacity relation with gas constant----- d) Helmholtz free energy (A) is defined as $A = H + TS$ (True/False) e) An isolated system permits exchange of energy with its surrounding (True/False)	CO1, CO2
b.	Answer the following: i. Write down the three adiabatic relations i.e. in terms of PV, TV, TP.? ii. What is Throttling process iii. Define extensive property? Is density an extensive property? iv. Define the efficiency of heat engine? v. Define saturated phase and saturation Pressure	CO1, CO2
c.	Discuss the two propositions involved in the Carnot principle? Discuss Ideal gas as the Carnot engine working substance, for the calculations efficiency of the engine?	CO3
Q2		
a.	Explain the three specifications for the quantitative definition of entropy? Calculate, the change in internal energy, change in enthalpy, work done and the heat supplied in an ideal gas contained in a vessel of 0.1 m ³ capacity is initially at 1 bar and 298 K. it is heated at constant volume to 400 K. Take $C_p = 30 \text{ J/mol K}$.	CO3, CO4
b.	a. Define polytropic process, what is the process called when value of $n=1$ and $n=\infty$? b. A reversible Heat engine receives heat from a high temperature reservoir at T_1 K and rejects heat to a low temperature sink at 600 °C. a second Reversible engine receives the heat rejected by the 1 st engine at 60 °C. Calculate the temperature T_1 , for equal thermal efficiencies of the two engine. c. Show that the Joule-Thomson coefficient is zero for ideal gas	CO1, CO4

c.	What do you mean by COP of a refrigerator? Establish relationship between COP of refrigerator and Efficiency of a heat engine. Solve: Mercury has a density of $13.69 \times 10^3 \text{ kg/m}^3$ in the liquid state and $14.193 \times 10^3 \text{ kg/m}^3$ in the solid state, both measured at the melting point of 234.33 K at 1 bar. If the heat of fusion of mercury is 9.7876 kJ/kg, what is the melting point of mercury at 10 bar?	CO1, CO5
Q3		
a.	Classify the thermodynamics properties of pure fluids? How Maxwell's equations are helpful? Derive the Maxwell's equations from the exact differential equations of the four energy properties.	CO1
b.	Establish the entropy-heat capacity relationships and hence: Determine the increase in entropy of solid magnesium when the temperature is increased from 300 K to 800 K at atmospheric pressure. The heat capacity is given by the equation as: $C_p = 26.04 + 5.586 \times 10^{-3}T + 28.476 \times 10^{-4}T^{-2}$	CO1, CO2
c.	What do you mean by the COP of heat pump? Hence solve: A house requires $2 \times 10^5 \text{ kJ/h}$ for heating in winter. Heat pump is used absorb heat from cold air outside in winter and send heat to the house. Work required to operate the heat pump is $3 \times 10^4 \text{ kJ/h}$. Determine: i. Heat absorbed from outside ii. Coefficient of Performance [COP]	CO2, CO5
Q4		
a.	Explain Gibbs free energy and hence Derive the modified equations for Internal energy and enthalpy by using the energy and Maxwell's relations? Specify the symbol used.	CO1, CO5
b.	With the help of curve explain the inversion point of Joule Thompson co-efficient (μ). Prove that the Joule Thompson co-efficient (μ) in Throttling Process can be expressed by the expression as: $\mu = \frac{[T(\frac{dV}{dT})_p - V]_{\text{constant } h}}{C_p}$	CO4, CO5
c.	Explain fugacity and fugacity coefficient and their SI unit? Derive the expression for the determination of fugacity of pure gases by using compressibility factor Z.	CO1, CO6
Q5		
a.	Define the following a. Partial molar properties b. Chemical potential c. Fugacity in solutions d. Effect of pressure on fugacity	CO4, CO6
b.	Discuss Chemical potential? Explain the effect of temperature and pressure on the chemical potential. Specify the symbol used.	CO4, CO5
c.	What do you mean by refrigerant? To design the refrigeration process the type of refrigerant is very important. What are the characteristics parameter on which the choice of refrigerant for the refrigeration process is to be considered? Discuss in details.	CO2, CO6